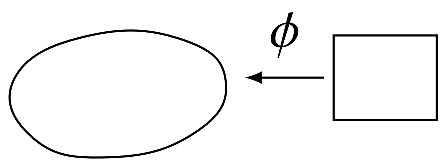
Chapter 1 — Abstract setting

Quadratic minimisation problems

MTH3340 · Numerical Methods for PDEs

1 Abstract setting

Minimise a functional J on a domain $\Omega \subseteq \mathbb{R}^d$, $\mathrm{diam}(\Omega) < \infty$, with $\partial\Omega$ piecewise smooth.



homeomorphism

- bijective
- continuous
- inverse continuous

Definition ((Bi)linear forms)

Given a vector space V on the field of scalars $\mathbb{R}$, a form $\ell : V \to \mathbb{R}$ is linear if it satisfies

$$\ell(u+v) = \ell(u) + \ell(v), \quad u, v \in V, \qquad \ell(\alpha u) = \alpha \ell(u) \quad \forall u \in V,\ \alpha \in \mathbb{R}.$$

A form $a : V \times V \to \mathbb{R}$ is a bilinear form if it is linear with respect to each of its two arguments separately.

Ex.: $\ell(v) = \int_\Omega f v$ is linear, $a(u, v) = \int_\Omega \partial u \cdot \partial v$ is bilinear (check it!)

2 Quadratic minimisation problem

Definition (Quadratic minimisation problem)

Let us consider a functional $J : V \to \mathbb{R}$ on a vector space V that can be expressed as

$$J(v) \doteq \frac{1}{2}a(v,v) - \ell(v) + c$$

for a symmetric bilinear form $a : V \times V \to \mathbb{R}$, a linear form $\ell : V \to \mathbb{R}$ and $c \in \mathbb{R}$. The problem $u = \mathrm{argmin}_{v \in V} J(v)$ is called a quadratic minimisation problem (QMP) on V.

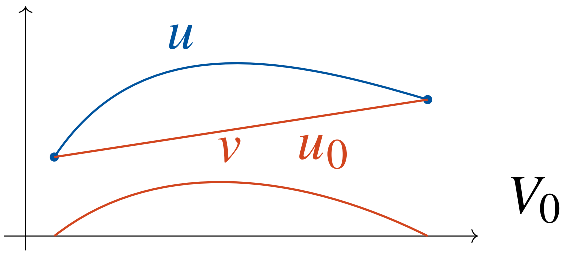
Is our model problem a QMP? a bilinear ✓ ℓ linear ✓ V a vector space ×

If we take $V \equiv \{u \in C^1([a,b]) : u(a) = u_a,\ u(b) = u_b\}$ and $u, v \in V$, then $u + v \notin V$ and $\alpha u \notin V$, due to the non-zero (non-homogeneous) bc's.

Offset function method

Pick an *arbitrary* $u_0 \in V$ (i.e., satisfying the bc's). We can express any $u \in V$ as $u_0 + v$ with $v \in C_0^1([a,b])$,

$$V \equiv u_0 + V_0, \qquad V_0 \doteq C_0^1([a,b]) \quad \text{this is a vector space!}$$



Thus, our problem is equivalent to

Equivalent problem on a vector space

$$u = u_0 + \arg\min_{v \in V_0} J(u_0 + v) \qquad \text{for an arbitrary } u_0 \in V.$$

Is this still a QMP?

$*$ Does $J(u_0 + v)$, as a functional of v, satisfy the definition of QMP? (u_0 is given, v is to be obtained by minimisation.)

If $J(v)$ satisfies the definition, $J(u_0 + v)$ does too,

$$\begin{aligned}
J(u_0 + v) &= \frac{1}{2}a(u_0 + v, u_0 + v) - \ell(u_0 + v) + c \\
&= \frac{1}{2}a(v, v) + a(u_0, v) + \frac{1}{2}a(u_0, u_0) - \ell(v) - \ell(u_0) + c \\
&= \frac{1}{2}a(v, v) - \underbrace{[\ell(v) - a(u_0, v)]}_{\tilde{\ell}(v)} + \underbrace{c + \frac{1}{2}a(u_0, u_0) - \ell(u_0)}_{\tilde{c}} \qquad \checkmark
\end{aligned}$$

$\tilde{\ell}$ is linear and $\tilde{c}$ is a constant, so we have a QMP on the vector space V_0.

Towards well-posedness

We need to impose some constraints on a, ℓ to prove well-posedness ($\exists!$ of a global minimum for the QMP).

Definition (Positive definiteness)

A symmetric bilinear form $a : V_0 \times V_0 \to \mathbb{R}$ on a real vector space V_0 is positive semi-definite if $a(u,u) \geq 0$ for all $u \in V_0$. Moreover, it is positive definite if $a(u,u) > 0$ for all $u \in V_0 \setminus \{0\}$.

Examples:

$$\int_\Omega uv \ \text{ is PD on } C^0([a,b]), \qquad \text{since } \int_\Omega u^2 = 0 \ \longrightarrow \ u \equiv 0.$$

$$\int_\Omega u'v' \ \text{ is semi-PD on } C^1([a,b]) \quad \Big(\int_\Omega (u')^2 = 0 \ \text{ for } u = \text{const.}\Big),$$

but it is PD on $C_0^1([a,b])$ (the only constant in this space is $u \equiv 0$).

Partial results (i)

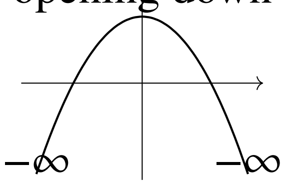
If the QMP has a solution $\longrightarrow$ a is positive semi-definite.

Lemma (Necessary condition for existence)

If the quadratic minimisation problem has a solution, then its bilinear form $a : V_0 \times V_0 \to \mathbb{R}$ must be semi-positive definite.

Proof. If a is not semi-positive definite, we can pick $u \in V_0$ such that $a(u,u) < 0$. Thus, $J(\alpha u) = \frac{1}{2}\alpha^2 a(u,u) - \alpha \ell(u) + c$ and $\lim_{\alpha \to \infty} J(\alpha u) = -\infty$. Thus, no global minimum. $\square$

$J(\alpha u)$, a parabola opening down



Partial results (ii)

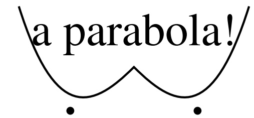
If the QMP has a solution (how do we know?) & a is PD $\longrightarrow$ the solution is unique.

Lemma (Sufficient condition for uniqueness)

If the quadratic minimisation problem has a solution and its bilinear form a is positive definite, then the solution is unique.

Proof. Assume two solutions $u \neq v$. Then $\Phi(\alpha) \doteq J(\alpha u + (1-\alpha)v)$ has two distinct global minima, at $\alpha = 0$ and $\alpha = 1$. Besides, $\Phi(\alpha) = \frac{\alpha^2}{2} a(u-v, u-v)$+lower order terms and $a(u-v, u-v) > 0$, so Φ is a non-degenerate parabola that opens up; it has its only minimum at its vertex. Contradiction. $\square$

two minima
not possible for
a parabola!



Norms and inner products

We need some additional notation to prove existence (uniqueness is then a consequence, by the previous lemma).

Definition (Norm on a vector space)

A norm $\|\cdot\|$ on a vector space V is a map $\|\cdot\| : V \longrightarrow \mathbb{R}_+$ such that

- positive definiteness, $\|v\| = 0 \iff v = 0$,
- absolute homogeneity, $\|\alpha v\| = |\alpha|\|v\|, \forall \alpha \in \mathbb{R}, \forall v \in V$,
- triangle inequality, $\|v + w\| \leq \|v\| + \|w\|, \forall v, w \in V$.

Energy norm and inner product

Definition (Energy norm)

A symmetric positive definite bilinear form $a : V \times V \to \mathbb{R}$ induces the energy norm $\|u\|_a \doteq a(u,u)^{1/2}$.

An inner product $(\cdot,\cdot)$ on V is a symmetric positive definite bilinear form on V.

E.g., $a(u,v) = \displaystyle\int_\Omega uv,$

$$\|u\|_a = \left[\int_\Omega u^2\right]^{1/2}.$$

$\int_\Omega uv$ is an inner product on C^0 functions.

Remember the Cauchy–Schwarz inequality, $(v,w) \le \|v\|\,\|w\|$ for $\|v\| \doteq (v,v)^{1/2}$. Same here for $a(\cdot,\cdot)$ and $\|\cdot\|_a$, $\quad a(u,v) \le \|u\|_a\|v\|_a$.

Continuity

Definition (Continuity of (bi)linear forms)

Given a vector space V with norm $\|\cdot\|$, a linear form $\ell : V \longrightarrow \mathbb{R}$ is continuous (or bounded) if

$$\exists\, C > 0 \ : \ |\ell(v)| \le C\|v\| \quad \forall v \in V.$$

A bilinear form $a : V \times V \longrightarrow \mathbb{R}$ is continuous if

$$\exists\, K > 0 \ : \ |a(u, v)| \le K\|u\|\,\|v\| \quad \forall u, v \in V.$$

Example: in our case,

$$\int_\Omega f v \le \Big(\int_\Omega f^2\Big)^{1/2}\Big(\int_\Omega v^2\Big)^{1/2} \quad \checkmark \qquad \text{for } \|v\|^2 = \int_\Omega v^2.$$

For the bilinear form, continuity holds with $K = 1$ if we pick $\|u\| = \|u\|_a$ (by Cauchy–Schwarz).

Coercivity

Continuity bounds a from above; coercivity bounds it from below.

Definition (Coercivity)

Given a vector space V with norm $\|\cdot\|$, a bilinear form $a : V \times V \to \mathbb{R}$ is *coercive* if

$$\exists\, \alpha > 0 \;:\; a(v, v) \geq \alpha \|v\|^2 \quad \forall v \in V.$$

A functional $J : V \to \mathbb{R}$ is *coercive* if $J(v) \to \infty$ as $\|v\| \to \infty$.

PD $\iff$ $\|\cdot\|_a$ is a norm — but possibly a *weaker* one! E.g., $a(u, v) = \int_0^1 uv$ with $\|v\|^2 = \int_0^1 v^2 + (v')^2$; take $v_n = \frac{1}{n}\sin(n\pi x)$; not coercive.

Coercivity + continuity $\iff$ the two norms are equivalent,

$$\alpha\|v\|^2 \leq a(v, v) = \|v\|_a^2 \leq K\|v\|^2 \qquad (\alpha = K = 1 \text{ for } \|\cdot\| = \|\cdot\|_a).$$

A genuine requirement when the norm is fixed beforehand (next chapters!).

Boundedness from below

a PD, J bounded below $\iff$ ℓ is continuous w.r.t. $\|\cdot\|_a$, i.e., $\ell(v) \le C\|v\|_a$.

Lemma (Boundedness from below)

The quadratic functional J with a positive definite bilinear form a is bounded from below on V_0 if and only if the linear form ℓ is continuous on V_0 with respect to the energy norm $\|\cdot\|_a$.

Proof. ($\Leftarrow$) If a is positive definite and ℓ is continuous, using the generalised Young's inequality $ab \le \frac{a^2}{2\epsilon} + \frac{\epsilon b^2}{2}$, $a, b \in \mathbb{R}$, $\epsilon > 0$, we readily get

$$J(u) = \frac{1}{2}a(u,u) - \ell(u) \ge \frac{1}{2}\|u\|_a^2 - C\|u\|_a \ge -\frac{1}{2}C^2.$$

($\Rightarrow$) Assume J bounded below and ℓ not continuous. For every $n \in \mathbb{N}$, we can pick $u_n \in V_0$ such that $\ell(u_n) \ge n\|u_n\|_a$. Re-scaling $\tilde{u}_n \leftarrow u_n/\|u_n\|_a$, we can assume $\|\tilde{u}_n\|_a = 1$, thus

$$J(\tilde{u}_n) \le \tfrac{1}{2} - n \to -\infty, \quad \text{as } n \to \infty,$$

so J cannot be bounded below. Contradiction. $\square$

ℓ must be continuous and a PD for $\exists!$.

Existence and uniqueness in finite dimension

Still, we need more properties for $\exists!$ on an ∞-dimensional V_0. In finite dimension, we have all we need.

Theorem (Existence and uniqueness of a minimiser in finite dimension)

If the vector space V_0 in the quadratic minimisation problem has finite dimension, a is a positive definite symmetric bilinear form and ℓ is a continuous linear form, then there exists a unique solution of this problem.

Since $\dim V_0 < \infty$, take a basis, $V_0 = \operatorname{span}\{\phi_1, \dots, \phi_N\}$. a is an inner product: orthonormalise the basis wrt a (Gram–Schmidt), $a(\phi_i, \phi_j) = \delta_{ij}$; then $\|v\|_a = |\vec{v}|$ (isometric isomorphism $V_0 \cong \mathbb{R}^N$). At the discrete level,

$$J(v) = \frac{1}{2}\vec{v} \cdot \vec{v} - \vec{f} \cdot \vec{v} + c, \qquad \vec{f}_i \doteq \ell(\phi_i),$$

a paraboloid opening up in $\mathbb{R}^N$; its unique stationary point (the vertex; as for the model problem, formalised next lecture) is

$$\vec{u} = \vec{f}, \qquad \text{i.e.,} \qquad u = \textstyle\sum_i \ell(\phi_i)\, \phi_i \qquad \exists! \checkmark$$

A more geometric route: compactness

The proof above is *pure linear algebra* (a basis, Gram–Schmidt, an explicit formula). What will survive in infinite dimensions? Let us redo it geometrically.

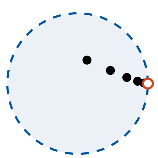
Definition (Compact set)

A set $K \subset V$ is *compact* if every sequence in K has a subsequence converging to a limit that *belongs to* K.

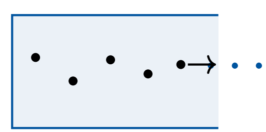
Compactness provides clustering for bounded sequences.

Theorem (Heine–Borel theorem)

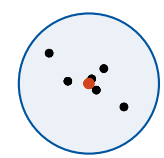
A set $K \subset \mathbb{R}^d$ is compact if and only if it is closed and bounded. (Easy!)



bounded, not closed:
the limit escapes



closed, not bounded:
sequences run away



closed and bounded = compact:
a subsequence converges inside

Alternative proof, via compactness

Theorem (Weierstrass extreme value theorem)

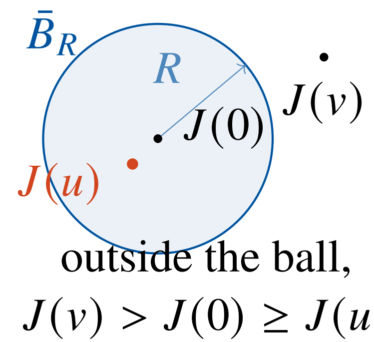
A continuous function $f : K \to \mathbb{R}$ on a compact set K attains its minimum and its maximum on K.

Alternative proof (sketch). Take the a-orthonormal basis $\{\phi_i\}_{i=1}^N$ above: $\|v\|_a = |\vec{v}|$, so we work with the representatives $\vec{v} \in \mathbb{R}^N$. J is a quadratic polynomial in $\vec{v}$, thus continuous; ℓ continuous reads $|\ell(v)| \le C\|v\|_a = C|\vec{v}|$, so

$$J(v) = \tfrac{1}{2}|\vec{v}|^2 - \ell(v) + c \ge \tfrac{1}{2}|\vec{v}|^2 - C|\vec{v}| + c \to \infty \text{ as } |\vec{v}| \to \infty$$

(J is coercive). Thus, for R large enough, $J(v) > J(0)$ for $|\vec{v}| > R$: minimisation happens in the ball $\bar{B}_R \doteq \{\vec{v} \in \mathbb{R}^N : |\vec{v}| \le R\}$, closed and bounded, i.e., compact by Heine–Borel. By Weierstrass, J attains its minimum over $\bar{B}_R$ at some $\vec{u}$, i.e., at $u \doteq \sum_i \vec{u}_i \phi_i \in V_0$. It is global: for $|\vec{v}| > R$, $J(v) > J(0) \ge J(u)$. □

Clustering allows us to prove the Weierstrass theorem (we skip the proof).



Compactness converts boundedness into existence.

Where this breaks

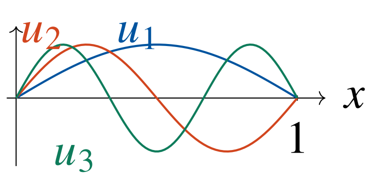
The sphere (closed unit ball) IS COMPACT in $\mathbb{R}^d$ (Heine–Borel) — GOING TO BREAK in ∞ dim!

In pure linear algebra terms, consider the vector space

$$V \doteq \operatorname{span}\{\sin(n\pi x) \ : \ n = 1, 2, \ldots\}, \qquad \|v\| \doteq \left(\int_0^1 v^2\right)^{1/2}.$$

A basis with infinitely many elements: an ∞-dimensional space! For the normalised basis vectors $u_n(x) \doteq \sqrt{2}\,\sin(n\pi x)$,

$$\|u_n\| = 1 \ \forall n, \qquad \|u_n - u_m\|^2 = 2 \ \text{ for } n \neq m \quad \text{(check it!)}$$



Just like the e_i of $\mathbb{R}^d$! But there a sequence of basis vectors must *repeat* some e_i (finitely many!) $\to$ convergent subsequence. Here $u_1, u_2, \ldots$ never clusters: the unit ball is not compact.

No compactness $\to$ no Weierstrass: we need a *different mechanism* $\longrightarrow$ completeness (next lectures!).

The wrong space breaks existence

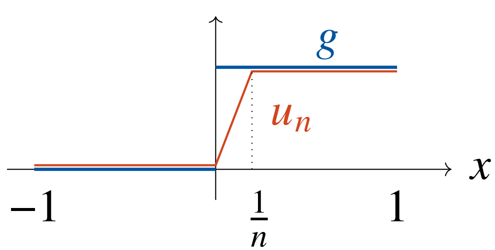
If we define the problem in the wrong space, we won't have ∃.

Example (Non-existence for a positive definite QMP)

Take $V_0 \doteq C^0([-1,1])$ and the *discontinuous* target $g \doteq 0$ on $[-1,0]$, $g \doteq 1$ on $(0,1]$, i.e., the PD QMP with $a(u,v) \doteq \int_{-1}^{1} uv$, $\ell(v) \doteq \int_{-1}^{1} gv$:

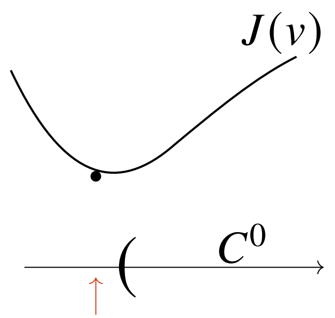
$$J(u) = \int_{-1}^{1} \tfrac{1}{2}u^2 - gu \,\mathrm{d}x = \tfrac{1}{2}\int_{-1}^{1}(u-g)^2\,\mathrm{d}x - \tfrac{1}{2} \geq -\tfrac{1}{2}.$$

The continuous ramps u_n (figure) give $J(u_n) = \frac{1}{6n} - \frac{1}{2} \to -\frac{1}{2}$ (check it!), so $\inf J = -\frac{1}{2}$. But $J(u) = -\frac{1}{2}$ would force $\int_{-1}^{0} u^2 = \int_{0}^{1}(u-1)^2 = 0$, i.e., $u \equiv 0$ on $[-1,0)$ AND $u \equiv 1$ on $(0,1]$ — impossible at $x = 0$ by continuity! (fine for a discontinuous function: g itself!) No minimiser!



J penalises the distance to g, and a continuous u cannot match the jump. The minimising sequence sharpens its layer and escapes the space: $u_n \to g \notin C^0([-1,1])$.

What went wrong?



the global min (g) does not belong to our space

Analogy: $f(x) = x^2$ has no global min in $(0, +\infty)$.

"Note: our space is not *closed*."

The space and the norm are mismatched; the energy norm is too weak for C^0 (its unit ball is not compact — previous board!).

Two possible fixes:

- a *stronger norm*? Not ours to choose — the energy norm is determined by the problem itself.
- enlarge the space so that limits of sequences cannot leave it, i.e., add the closure of the space wrt the a-norm. We must "complete" the space (next lectures!).